
Data Reduction

Himanshu Bijja
23311A05P5
Cse-D

Data Reduction Strategies

- **Need for data reduction**
 - A database/data warehouse may store terabytes of data
 - Complex data analysis/mining may take a very long time to run on the complete data set
- **Data reduction**
 - Obtain a reduced representation of the data set that is much smaller in volume but yet produce the same (or almost the same) analytical results
- **Data reduction strategies**
 - Data cube aggregation
 - Attribute Subset Selection
 - Numerosity reduction — e.g., fit data into models
 - Dimensionality reduction - Data Compression
 - Discretization and concept hierarchy generation

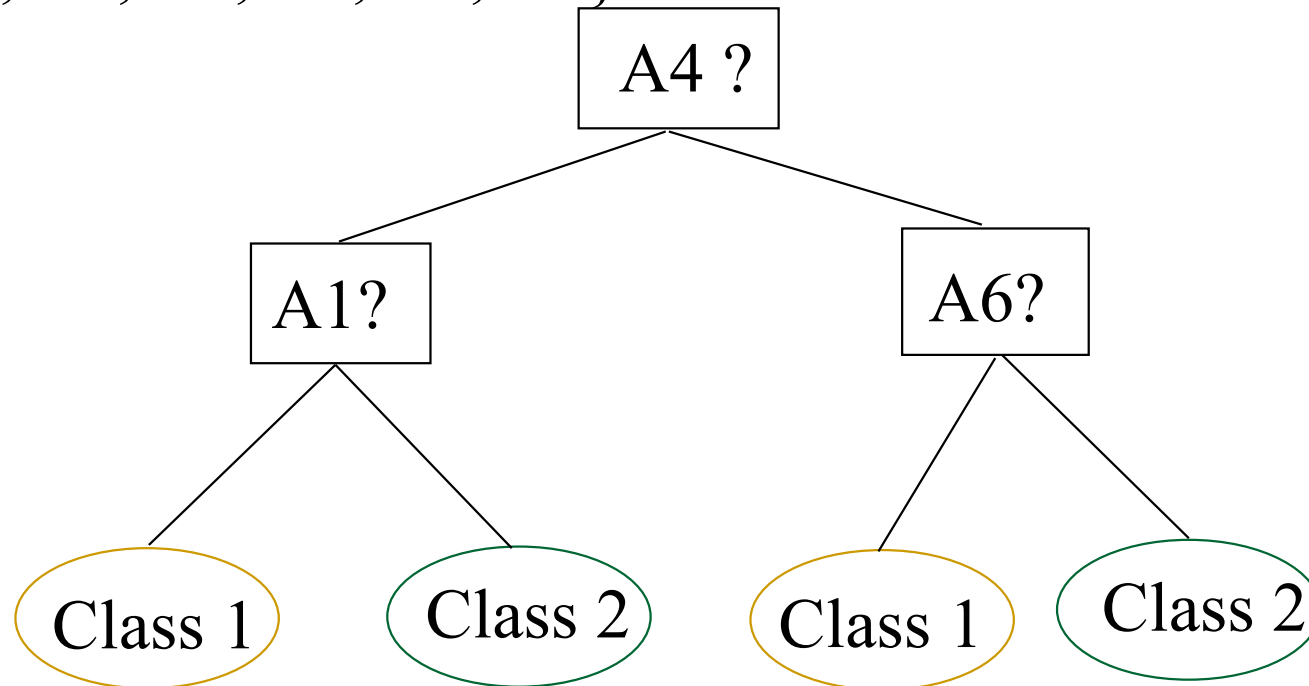
2. Attribute Subset Selection

- Feature selection (i.e., attribute subset selection):
 - Select a minimum set of features such that the probability distribution of different classes given the values for those features is as close as possible to the original distribution given the values of all features
 - reduce # of patterns - easier to understand
- Heuristic methods (due to exponential # of choices):
 - Step-wise forward selection
 - Step-wise backward elimination
 - Combining forward selection and backward elimination
 - Decision-tree induction

Example of Decision Tree Induction

Initial attribute set:

$\{A1, A2, A3, A4, A5, A6\}$



-----> Reduced attribute set: $\{A1, A4, A6\}$

Heuristic Feature Selection Methods

- There are 2^d possible sub-features of d features
- Several heuristic feature selection methods:
 - **Best single features** under the feature independence assumption:
choose by significance tests
 - **Best step-wise feature selection**
 - The best single-feature is picked first
 - Then next best feature condition to the first, ...
 - **Step-wise feature elimination**
 - Repeatedly eliminate the worst feature
 - Best **combined feature selection and elimination**

3. Numerosity Reduction

- Reduce data volume by choosing alternative, smaller forms of data representation
- Parametric methods
 - Assume the data fits some model, estimate model parameters, store only the parameters, and discard the data (except possible outliers)
 - Example: Log-linear models, Regression
- Non-parametric methods
 - Do not assume models
 - Major families: histograms, clustering, sampling

Regression and Log-Linear Models

- Regression

- Handles skewed data
- Computationally Intensive

- Log linear models

- Scalability
- Estimate the probability of each point in a multi-dimensional space based on a smaller subset
- Can construct higher dimensional spaces from lower dimensional ones

- Both

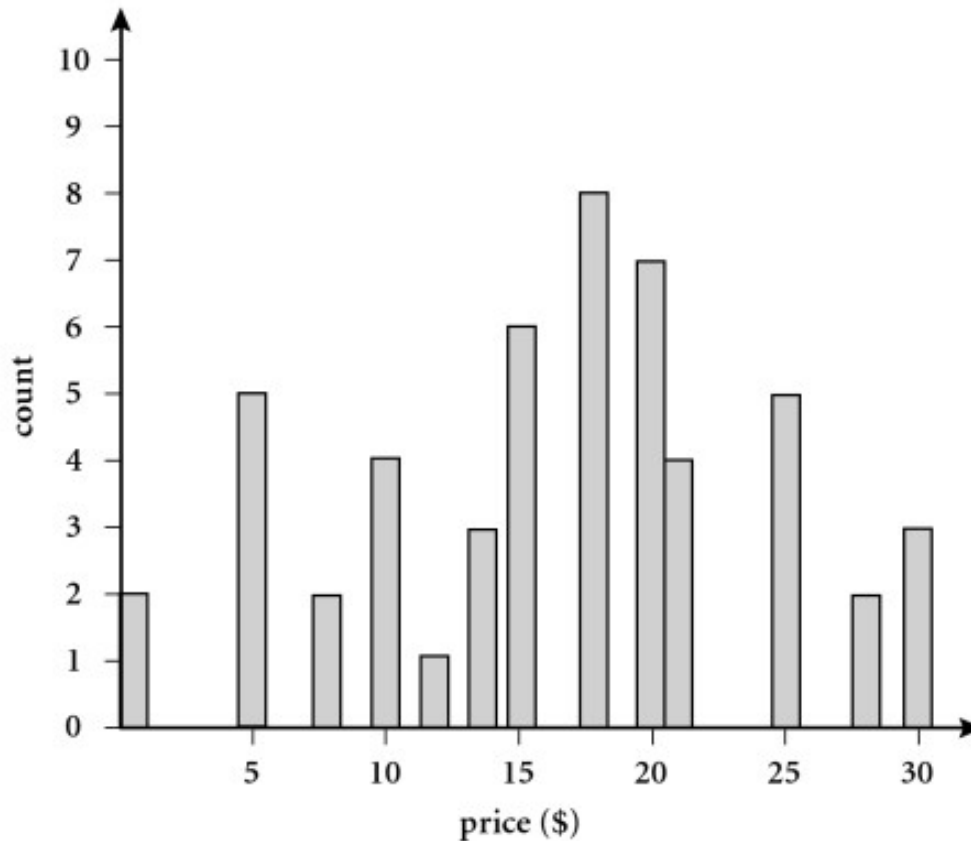
- Sparse data

Histograms

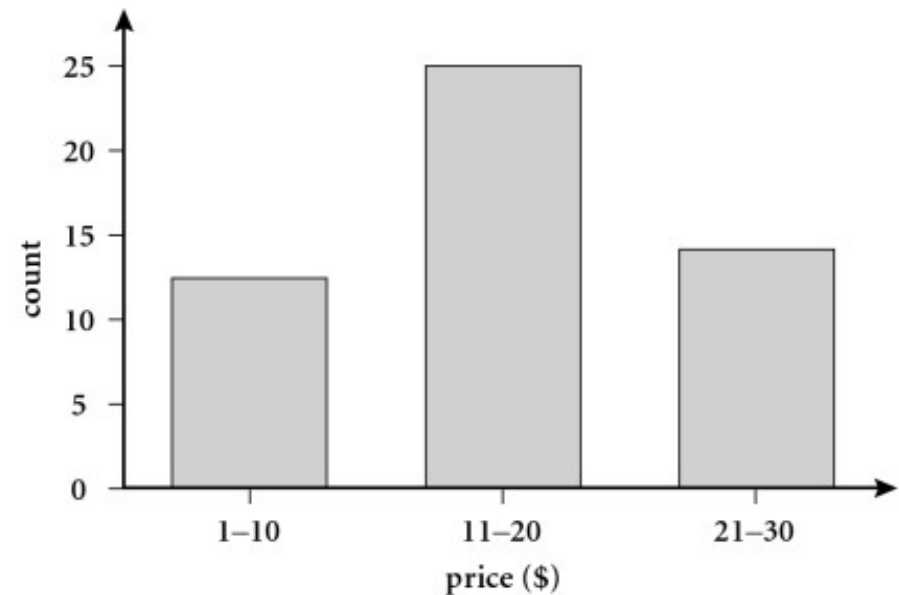
- Divide data into buckets and store average (sum) for each bucket
- Partitioning rules:
 - Equal-width: equal bucket range
 - Equal-frequency (or equal-depth)
 - V-optimal: with the least *histogram variance* (weighted sum of the original values that each bucket represents)
 - MaxDiff: Consider difference between pair of adjacent values. Set bucket boundary between each pair for pairs having the β (No. of buckets)–1 largest differences
- Multi-dimensional histogram

Histograms

Histogram with Singleton buckets



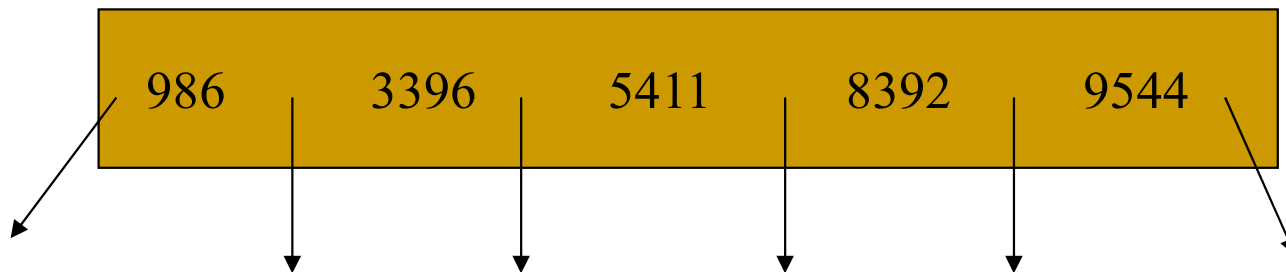
Equal-width Histogram



List of prices: 1, 1, 5, 5, 5, 5, 5, 8, 8, 10, 10, 10, 10, 12, 14, 14, 14, 15, 15, 15, 15, 15, 15, 18, 18, 18, 18, 18, 18, 20, 20, 20, 20, 20, 20, 20, 20, 21, 21, 21, 21, 25, 25, 25, 25, 25, 28, 28, 30, 30, 30

Clustering

- Partition data set into clusters based on similarity, and store cluster representation (e.g., centroid and diameter) only
- Can be very effective if data is clustered but not if data is “smeared”
- Can have hierarchical clustering and be stored in multi-dimensional index tree structures
 - Index tree / B+ tree – Hierarchical Histogram



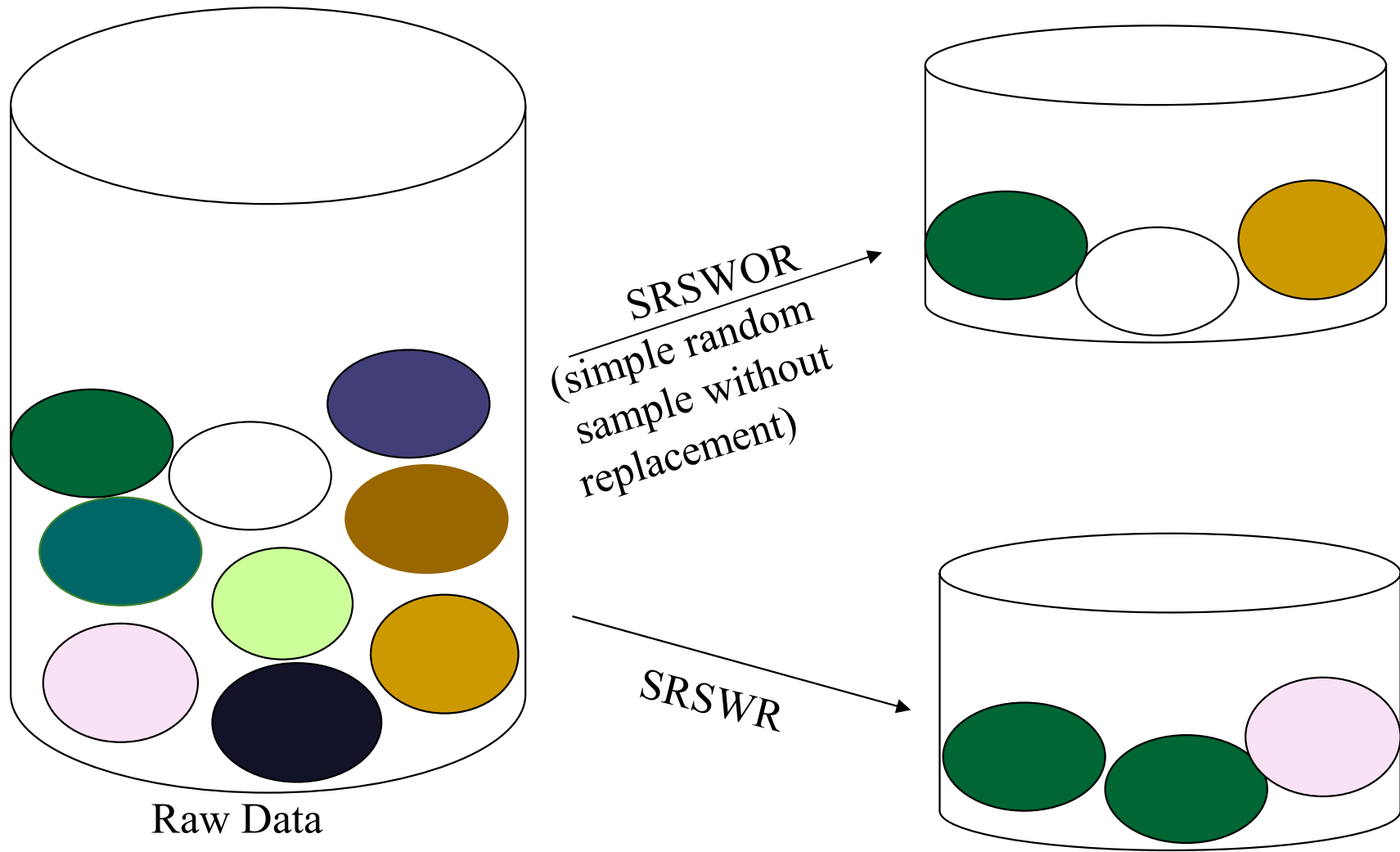
Sampling

- **Sampling:** obtaining a small sample s to represent the whole data set N
- Allow a mining algorithm to run in complexity that is potentially sub-linear to the size of the data
- Choose a **representative** subset of the data
 - Simple random sampling may have very poor performance in the presence of skew
- Develop adaptive sampling methods
 - Stratified sampling

Sampling Techniques

- Simple Random Sample **Without Replacement** (SRSWOR)
- Simple Random Sample **With Replacement** (SRSWR)
- Cluster Sample
- Stratified Sample

Sampling: with or without Replacement



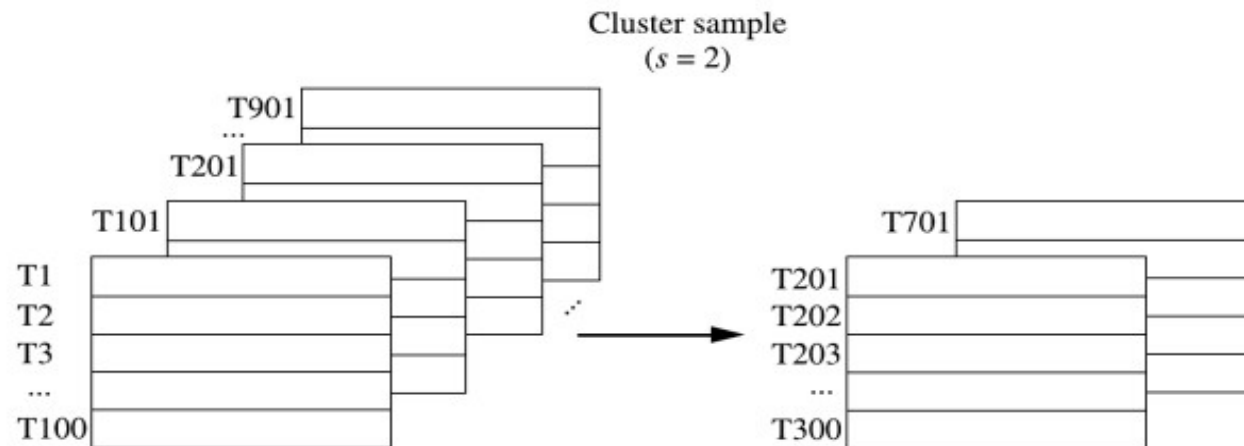
Cluster Sample

- Tuples are grouped into M mutually disjoint clusters
- SRS of m clusters is taken where $m < M$
- Tuples in a database retrieved in pages
 - Page - Cluster
 - SRSWOR to pages

Stratified Sample

- Data is divided into mutually disjoint parts called **strata**
- SRS at each stratum
- **Representative samples ensured** even in the presence of skewed data

Cluster and Stratified Sampling



Stratified sample
(according to *age*)

T38	youth
T256	youth
T307	youth
T391	youth
T96	middle_aged
T117	middle_aged
T138	middle_aged
T263	middle_aged
T290	middle_aged
T308	middle_aged
T326	middle_aged
T387	middle_aged
T69	senior
T284	senior

T38	youth
T391	youth
T117	middle_aged
T138	middle_aged
T290	middle_aged
T326	middle_aged
T69	senior

Features

- Cost depends on size of sample
- Sub-linear on size of data
- Linear with respect to dimensions
- Estimates answer to an aggregate query

Data Compression

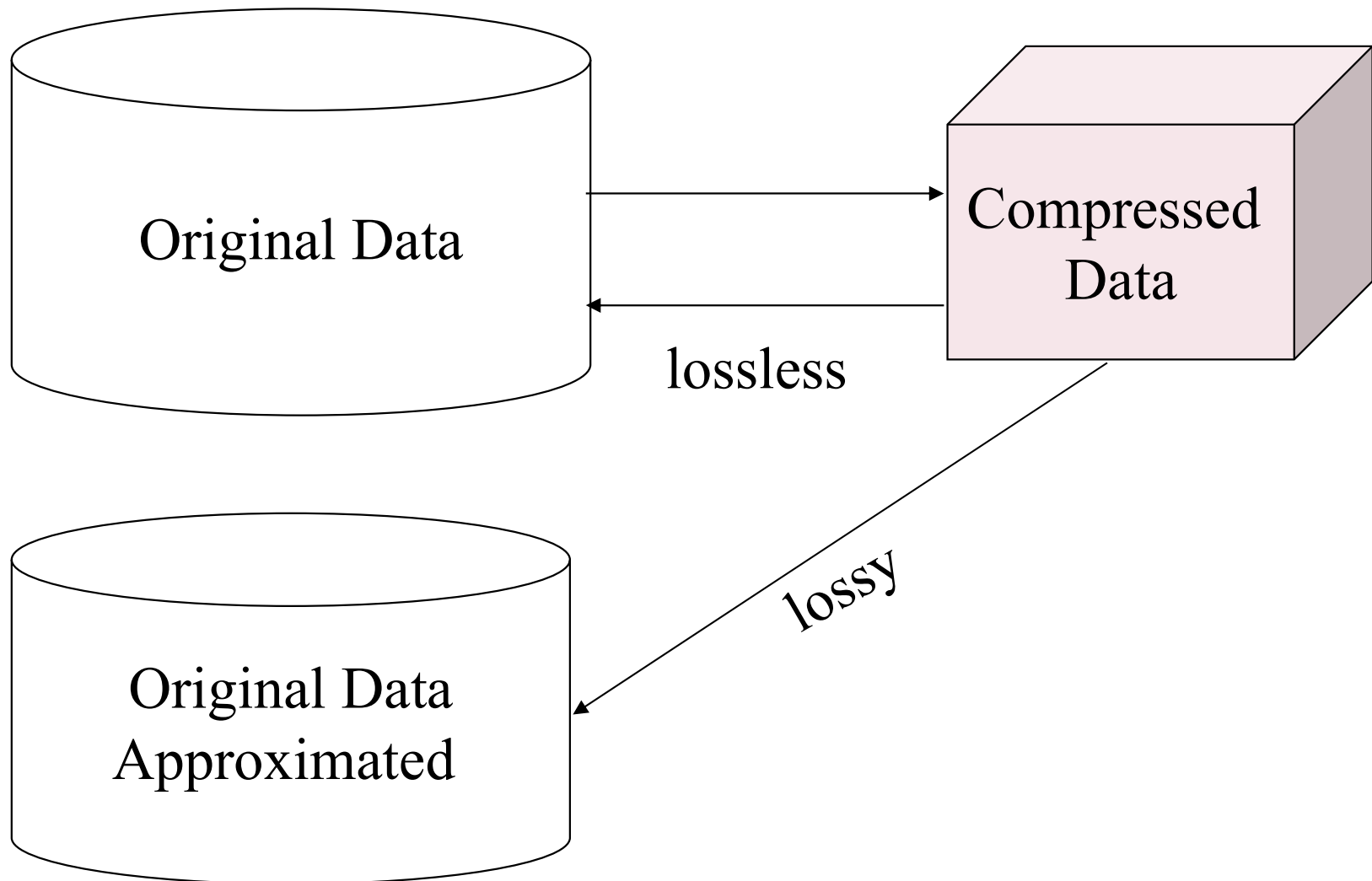
■ String compression

- There are extensive theories and well-tuned algorithms
- Typically **lossless**
- But only limited manipulation is possible

■ Audio/video compression

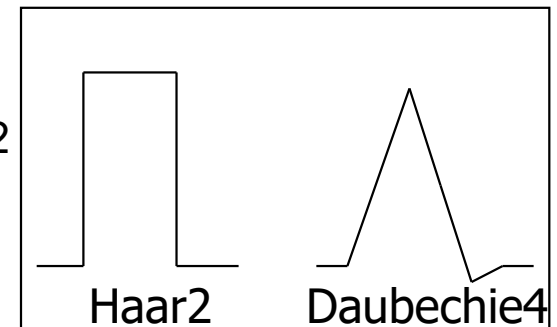
- Typically **lossy compression**, with progressive refinement
- Sometimes small fragments of signal can be reconstructed without reconstructing the whole

Data Compression



Dimensionality Reduction: Wavelet Transformation

- Discrete wavelet transform (DWT): linear signal processing
- Compressed approximation: store only a small fraction of the strongest of the wavelet coefficients
- Similar to discrete Fourier transform (DFT), but better lossy compression, localized in space
- Method:
 - Length, L , must be an integer power of 2 (padding with 0's, when necessary)
 - Each transform has 2 functions: smoothing, difference
 - Applies to pairs of data, resulting in two set of data of length $L/2$
 - Applies two functions recursively



Discrete Wavelet Transform

- Can also apply Matrix Multiplication
- Fast DWT algorithm
- Can be applied to Multi-Dimensional data such as Data Cubes
- Wavelet transforms work well on sparse or skewed data
- Used in Computer Vision, Compression of finger print images

Dimensionality Reduction: Principal Component Analysis (PCA)

- Given N data vectors from k -dimensions, find $c \leq k$ orthogonal vectors (*Principal components*) that can be best used to represent data
- Steps
 - Normalize input data: Each attribute falls within the same range
 - Compute c orthonormal (unit) vectors, i.e., *principal components*
 - Each input data (vector) is a linear combination of the c principal component vectors
 - The principal components are sorted in order of decreasing “significance” or strength
 - Since the components are sorted, the size of the data can be reduced by eliminating the weak components, i.e., those with low variance. (i.e., using the strongest principal components, it is possible to reconstruct a good approximation of the original data)
- Works for numeric data only
- Used for handling sparse data

Principal Component Analysis

